Mechanical Design

The Adaptive Wearable and Drone are designed using a **cost-effective, modular fabrication approach**. Rather than manufacturing the watch enclosure and other standard components from the ground up, we selected **commercially available watch housings and electronic components** that can accommodate our required hardware. This reduces fabrication time, material waste, and overall production cost while allowing the prototype to be assembled and tested more efficiently.

The electronic components are integrated into the selected enclosure and connected using **insulated wires and soldered electrical connections**. Soldering provides secure and reliable electrical connections between the sensors, microcontroller, display, LEDs, buzzer, and other components. Components that require physical mounting are secured within the enclosure using suitable mechanical fastening methods to prevent movement during operation.

The design is **modular**, allowing individual electronic components to be replaced, tested, or upgraded without redesigning the entire wearable. This approach also supports future mass production because commercially available enclosure components can be retained while the internal electronics are adapted to different configurations.

CAD renders, circuit diagrams, and prototype images are included to show the expected physical appearance, component arrangement, and assembly of the Adaptive Wearable.